



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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Computer Networking

OSI → ISO

Open
System Interconnect

International Organization
for Standardization

1984



Q. What is OSI Model?

- **OSI (Open Systems Interconnection) Model** is a **7-layer reference model** developed by the **ISO (International Organization for Standardization)** in 1984 to standardize and understand how data is communicated between computers over a network.
- The OSI model is now considered the primary Architectural model for inter-computer communications.

- This separation into smaller more manageable functions is known as layering.
- All these 7 layers work collaboratively/help to transmit the data from one person to another across/over the globe/world network.
- Each layer performs a **specific function**, and all seven layers work together to enable network communication.
- The OSI model is not an actual data communication protocol. It is a reference/conceptual model used to understand networking and standardize network communication.

7 Layers of OSI Model

7. Application

6. Presentation

5. Session

4. Transport

3. Network

2. Data Link

1. Physical

History of OSI Model (short)

OSI → Open Systems Interconnection

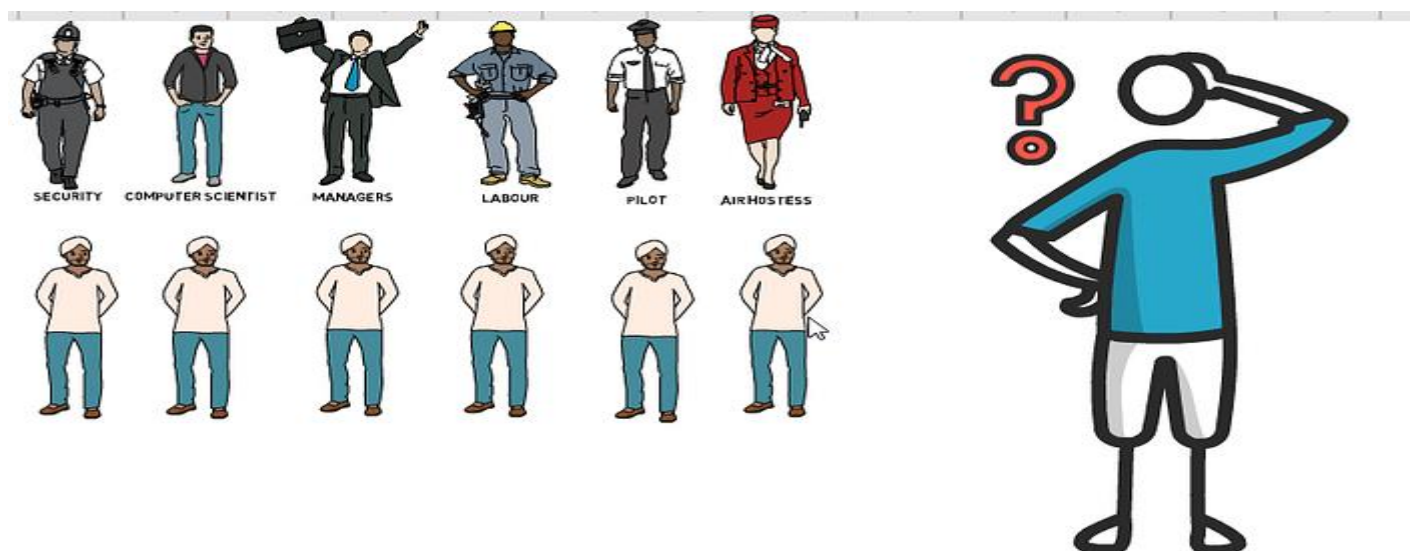
Developed by → **ISO**

Work started → **1977**

Seven-layer model standardized/published → **1984**

Number of layers → **7**

Standard → **ISO 7498**



Q. Why need OSI Model ?

The OSI model is needed to divide complex network communication into manageable layers, provide standardization, and make network design, understanding, and troubleshooting easier.

1. To simplify network communication

→ It divides complex communication into **7 manageable layers**.

2. To provide standardization

→ It provides a common framework for understanding how different network devices and protocols communicate.

3. To make troubleshooting easier

→ Network problems can be identified at a specific layer.

Example: Cable problem → **Physical Layer**; IP problem → **Network Layer**.

4. To understand network communication

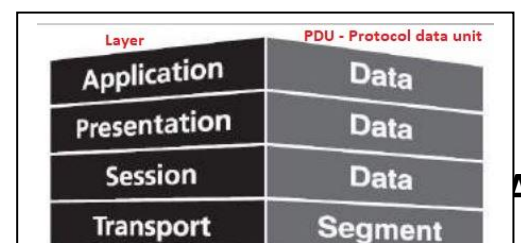
→ It helps us understand how data travels from the sender to the receiver.

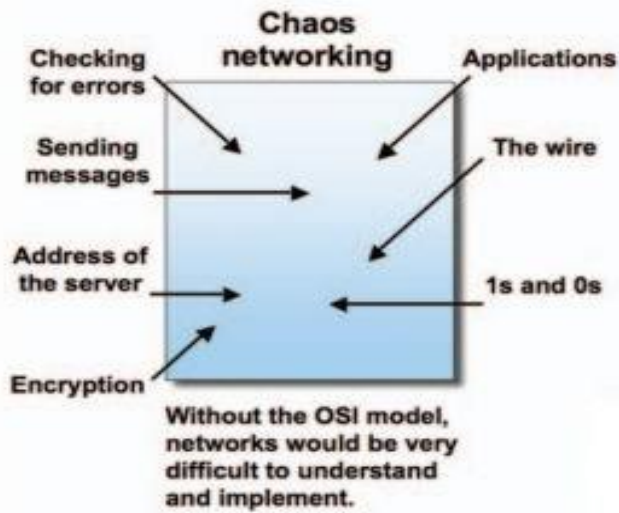
5. To help in network design and development

→ Each layer has specific responsibilities, making network technologies easier to design and develop.

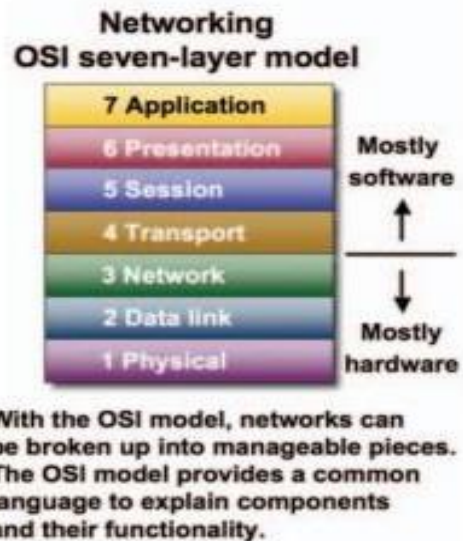
Q. What would happen if the OSI model did not provide standardization?

Without standardization, different manufacturers might use different networking methods and protocols, making it difficult for devices from different vendors to communicate and work together.



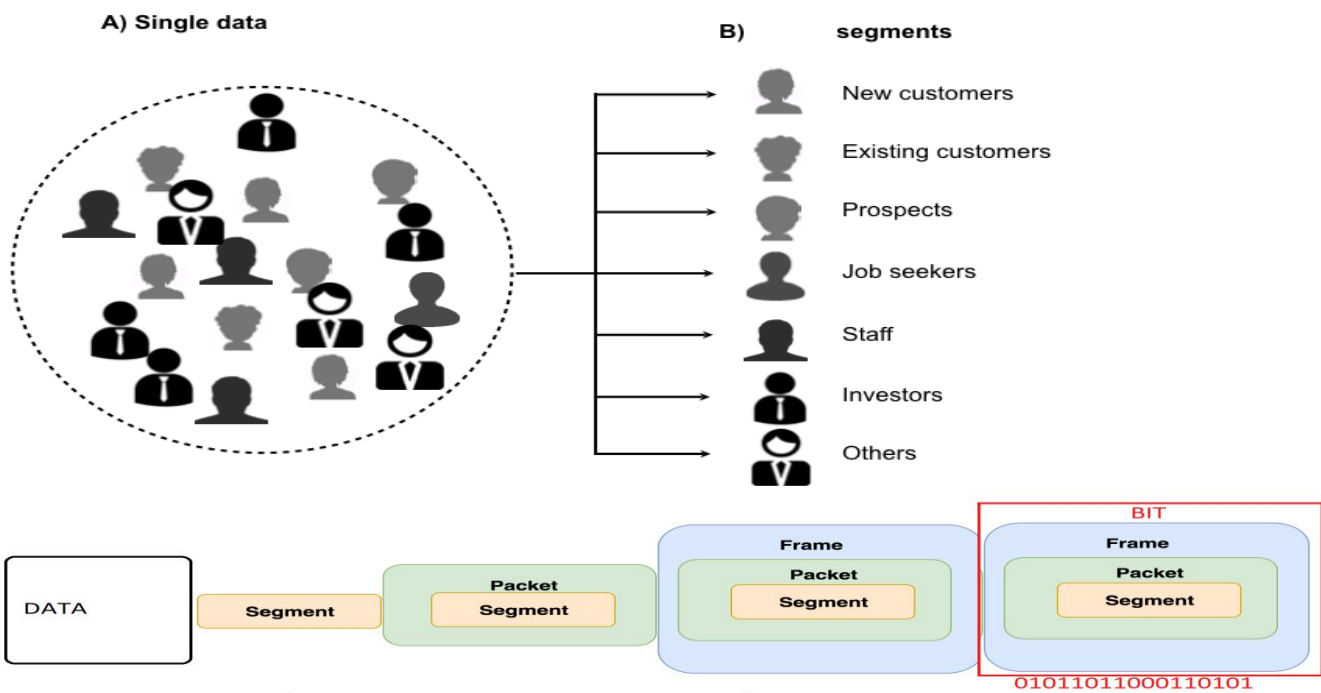


without OSI



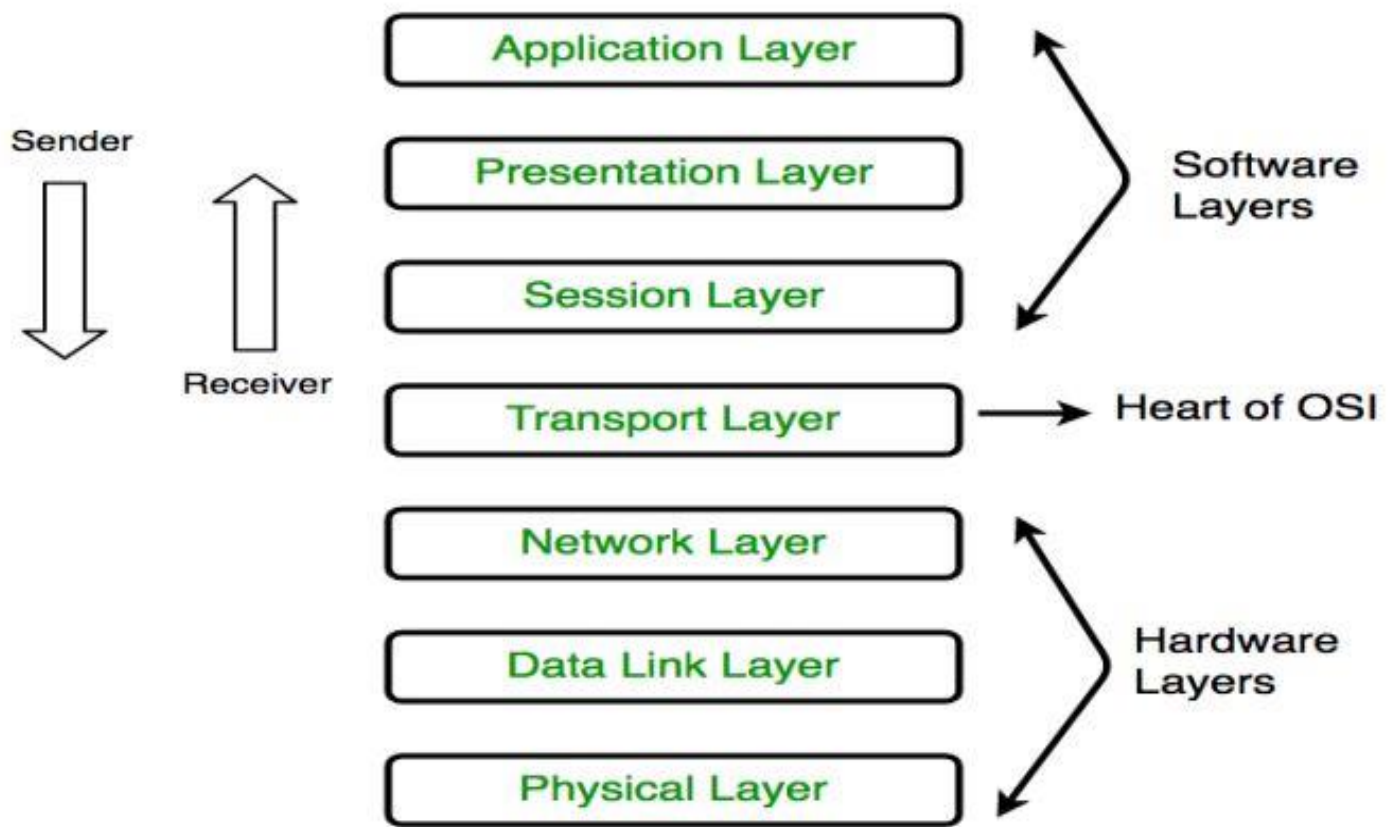
With OSI

Encapsulation	Data > segment > packet > frame > bits
De-encapsulation	Bits > frame > packet > segment > data



7 layer of OSI Model

All layers work together in the correct order to move data around a network.



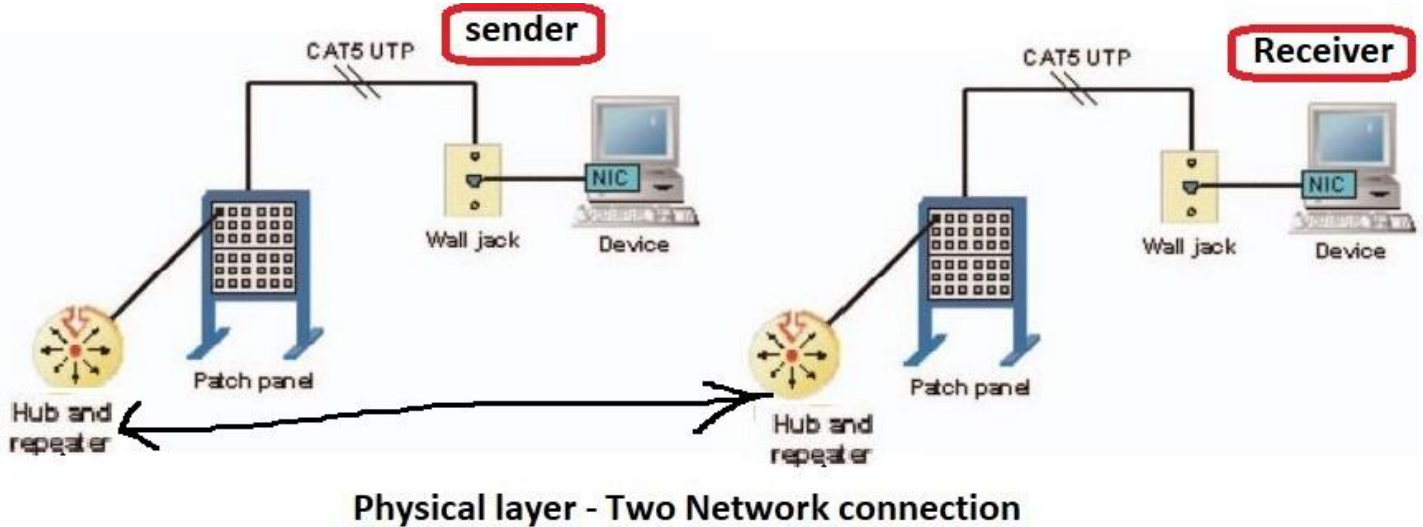
Two types of layer in OSI Model

Lower layers/Hardware layers	Higher layers/software layers
Layer 1 to Layer 4 (Lower layers) Layer 1 to Layer 3 (Hardware layers)	Layer 5 to Layer 7

1. Physical Layer

- First layer (Layer-1) of OSI model.
- Physical layer Used to for data transmit through Copper wiring, fiber optic cable, radio frequencies, anything that can be used to transmit data is defined on the Physical layer of the OSI Model.
- Physical layer of physically moving data (bit – 0/1) in signals from one computer to the next computer.
- The Physical Layer receives frame from the Data Link Layer and converts in bit and again bit transmits it as signals over the transmission medium.
Frame → Bits , Bits → signals
- Physical Layer defines the physical transmission of data over a communication medium(wire/wireless) and physical topology.

- Performs or defines analog-to-digital (A/D) and digital-to-analog (D/A) signal conversion for data transmission.
- **PDU** : bits
- **Device example**: Hub, repeater.

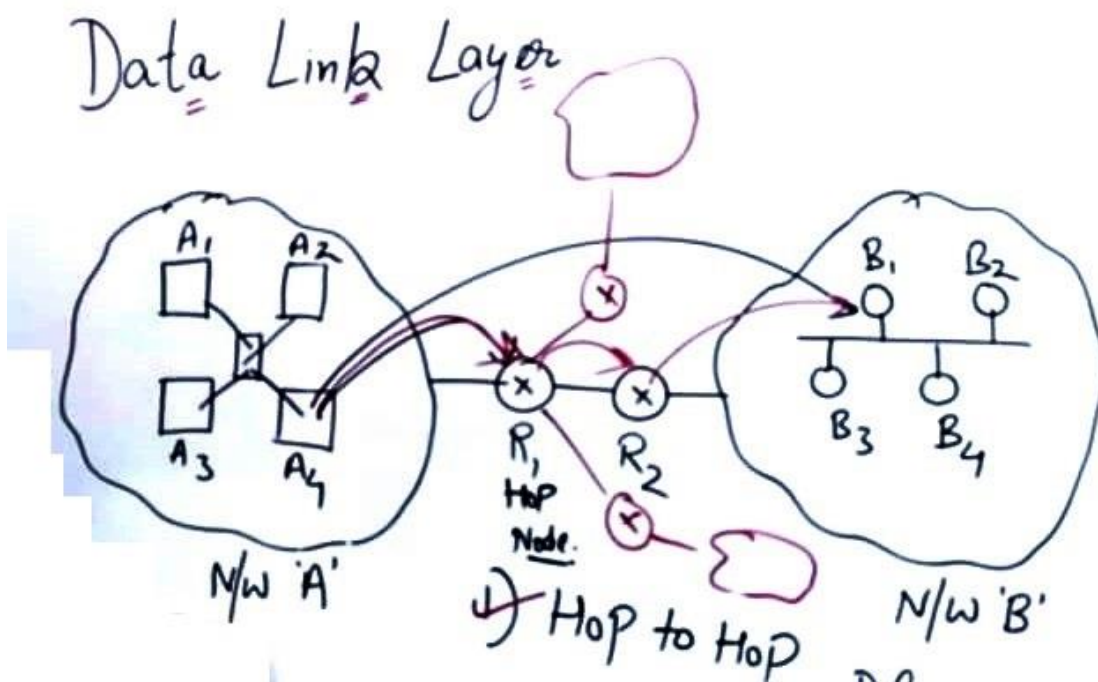
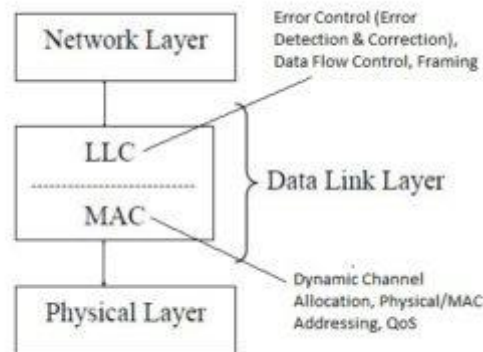


2. Data-Link Layer / Layer-2

- Second layer of OSI model.
- The Data Link Layer receives packets(data) from the Network Layer, encapsulates them into frames(data), and provides **node-to-node (hop-to-hop)** delivery through the Physical Layer and vice versa.
- Provides node-to-node communication within the same local network using MAC addresses. For communication between different networks, the Network Layer uses IP addresses for routing.
- Organizes data into frames by adding a header (put MAC) and trailer (put FCS/CRC), which help in addressing, error detection, and controlling access to the physical medium.
- A Data Link Layer frame contains source and destination MAC addresses (hardware/physical addresses), which are associated with the network interface (NIC).
- Frame = Header + packet(data) + Trailer
 - In which put, Header : source MAC + Destination MAC
 - Data : Packet of network layer
 - Trailer : FCS/CRC (for error detection)
- **2 Sub-layers of Data link layer**
 - I. **Logical Link Control (LLC)**: Manages flow control, error checking, and address notifications.

II. **Media Access Control (MAC):** Regulates physical medium access, frame boundaries, and hardware addressing.

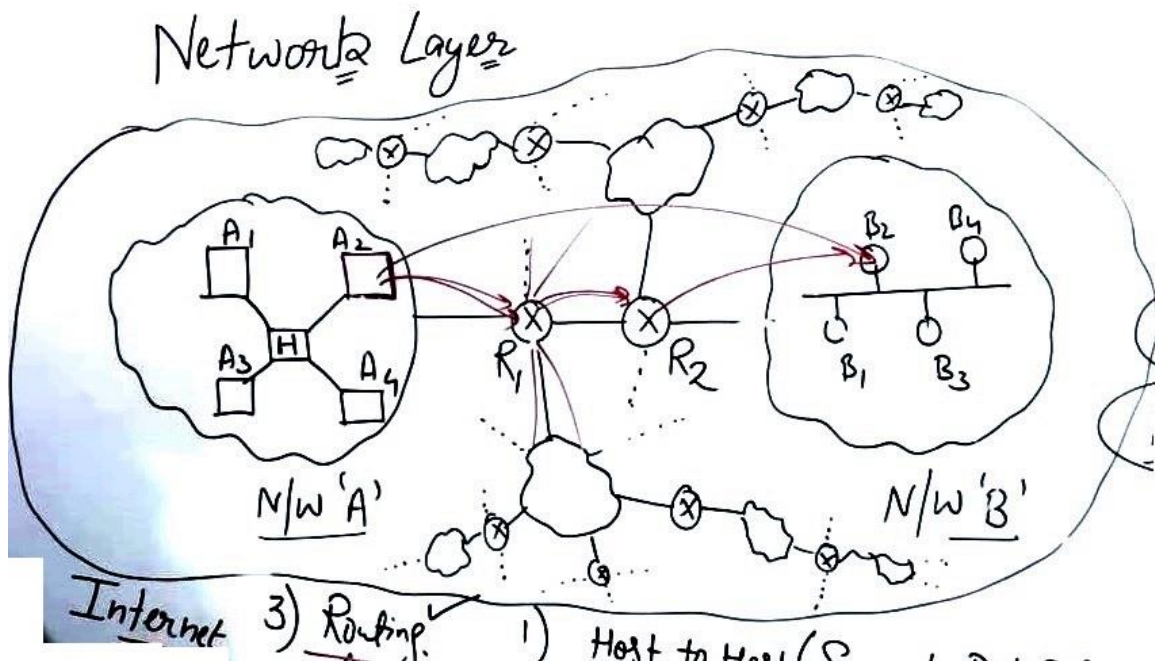
- Supports simplex, half-duplex or full-duplex communication.
- **PDU** : Frames
- **Physical address** / permanent : MAC address
- **Protocol** : ARP, PPP, HDLC(high level data link control), STP(spanning tree protocol)
- Data-link layer **Device example** :- Bridges & Switches.



3. **Network Layer / Layer-3**

- Third (3rd) layer of OSI model.
- The **Network Layer** receives data from the **Transport Layer** in the form of **segments** and converts them into **packets**.
- It is responsible for **source-to-destination (host-to-host)/communication** and determines the **best path/route** through the network.

- It uses **logical addressing**, such as **IP addresses**, to identify the source and destination networks and hosts.
- It determines **which network and which host** should receive the packet.
- **PDU: Packet**
- **Packet = IP Header + Segment** (In IP header contain - Source IP Address + Destination IP Address + Other IP Header Information) so, i.e. **Packet = Source IP Address + Destination IP Address + Other IP Header Information + Segment**.
- **Logical Address: IP Address**
- **Protocol : IP, ICMP, IGMP**
- **Network layer Device: Router**



4. Transport Layer / Layer-4

- Fourth(4th) layer of OSI model.
- The **Transport Layer** takes data from the higher layers of the OSI Model and breaks it into smaller **segments** (for TCP) that are passed to the Network Layer for transmission.
- It provides **End-to-End / Process-to-Process / Port-to-Port communication** between applications running on source and destination hosts.
- At the receiving side, the Transport Layer **reassembles the received segments** and delivers the data to the appropriate application.
- **Port-to-Port communication** means identifying **which**

application/process should receive the data. For example, when an email is sent using SMTP, a port number is used to identify the email service.

- It uses logical/software **port addressing**. Port numbers range from **0 to 65535**.
- Heart of OSI model is : Transport layer.
- PDU : Segment
- Segment = Transport Header (TCP/UDP) + Data (In transport header contain Source Port + Destination Port + Other Transport Header Information)
- Mainly two types of protocol use transport layer

(i) **TCP (Transmission Control Protocol)**

- ✓ Connection-oriented
- ✓ TCP, it is called a **TCP Segment**.
- ✓ Uses **three-way handshake** (SYN → SYN-ACK → ACK)
- ✓ Provides reliable and ordered delivery
- ✓ Provides retransmission/error recovery use checksum method
- ✓ Provides flow control
- ✓ Uses sequencing to maintain the correct order
- ✓ Examples: **HTTP/HTTPS, FTP, SMTP**

(ii) **UDP (User Datagram Protocol)**

- ✓ Connectionless
- ✓ UDP, it is called a **UDP Datagram**.
- ✓ Lightweight and fast
- ✓ Does not guarantee delivery
- ✓ Does not provide TCP-style retransmission and ordering
- ✓ Useful where speed and low delay are important
- ✓ Examples: **live streaming, online gaming, VoIP**

- **Multiplexing** : Multiplexing is the process in which data from **multiple applications/processes** is collected by the Transport Layer and sent through the network using different **port numbers (many to one)**.
- **Demultiplexing**: Demultiplexing is the reverse process. At the receiver side, the Transport Layer examines the **destination port number** and delivers the received data to the **correct application/process (one to many)**.

5. Session Layer / Layer 5

- Fifth(5th) layer of OSI model.
- The **Session Layer** is responsible for **establishing, managing, maintaining, and terminating sessions** between applications running on networked devices.
- It manages the **dialogue/conversation** between two applications and controls **who can transmit and when**.
- It provides **synchronization** between communicating applications. It can insert **checkpoints** into a communication session so that, if the session is interrupted, communication can potentially resume from a suitable checkpoint instead of starting over.
- **Example of checkpoint:**
Suppose a large file transfer is interrupted after 60%. If the application supports session checkpoints, it may resume from the saved point rather than starting from 0%.
- Session management can involve **authentication or login-related functions**, but authentication is not exclusively a Session Layer responsibility; in modern networks it can be implemented at several layers.
- The Session Layer is mainly a **logical/software functionality**. Its functions may be implemented through the **operating system, network stack, libraries, or application software**.

6. Presentation Layer / Layer 6

- Sixth(6th) layer of OSI model.
- Presentation Layer is also called the **Translation Layer**.
- It takes data from the Application Layer and converts it into a suitable format for communication.
- It performs three main tasks:
 - ✓ Translation / Code Conversion (ASCII to EBCDIC)
 - ✓ Compression / Decompression (.zip file)
 - ✓ Encryption / Decryption
- **Examples of data formats include:**
 - ✓ **Text:** ASCII, EBCDIC, RTF
 - ✓ **Images:** GIF, JPG, TIFF

- ✓ **Audio:** MIDI, MP3, WAV
- ✓ **Video:** MPEG, AVI, MOV
- **commonly associated protocols/standard of Presentation Layer:**
 - ✓ **TLS (Transport Layer Security)** — Encryption and decryption
 - ✓ **SSL (Secure Sockets Layer)** — Encryption and decryption
 - ✓ **XDR (External Data Representation)** — Data representation/encoding
- The presentation Layer is mainly a **logical/software functionality**. Its functions may be implemented through the **operating system, network stack, libraries, or application software**.

7. Application Layer / Layer 7

- Seventh(7th) layer of OSI model.
- The Application Layer (Layer 7) provides an interface between **user applications and network services**.
- It provides network services directly to applications used by the user.
- Examples of Application Layer protocols include **HTTP, HTTPS, FTP, SMTP, POP3, and DNS**.
- **Web Browser:** Firefox and Chrome are applications that use **HTTP/HTTPS** to communicate with web servers.
- **HTTP** stands for **Hyper Text Transfer Protocol**.
- **E-mail:** An e-mail program may use **POP3** to receive/read e-mails and **SMTP** to send e-mails.
- **POP3** stands for **Post Office Protocol version 3**.
- **SMTP** stands for **Simple Mail Transfer Protocol**.
- The Application Layer is the **topmost layer of the OSI model (Layer 7)**.

Q. What are the seven layers of the OSI model and explain their basic functions?

The seven layers of the OSI model are:

1. Physical Layer:

It is responsible for transmitting and receiving **raw bits** over a physical medium such as cables, wires, fiber-optic cables, or wireless signals.

2. Data Link Layer:

It is responsible for framing, physical (MAC) addressing, error detection, and flow control. It provides communication between devices on the same network link.

3. Network Layer:

It is responsible for logical addressing (IP addressing), routing, and forwarding of packets from the source to the destination across different networks.

4. Transport Layer:

It provides end-to-end (process-to-process) communication. Its main functions include segmentation, port addressing, flow control, error control, and connection management.

TCP provides reliable and ordered delivery, whereas **UDP** provides connectionless communication without guaranteeing delivery.

5. Session Layer:

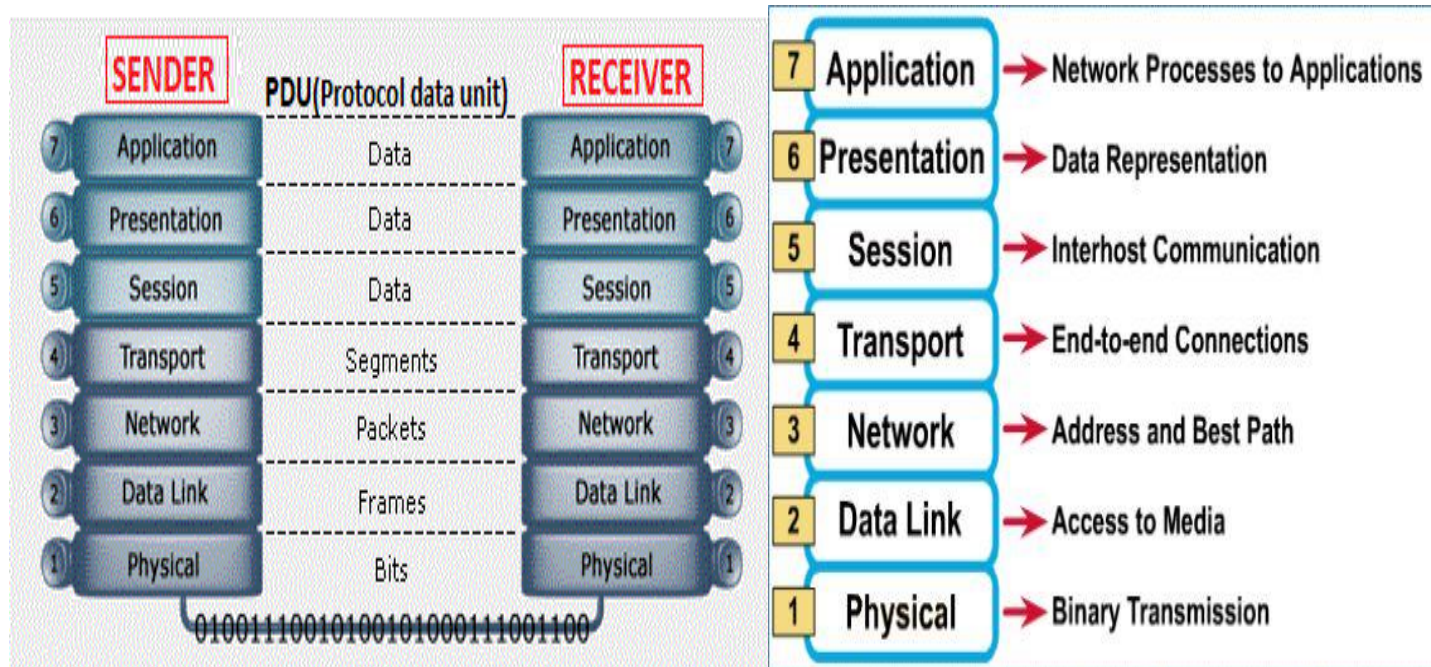
It is responsible for establishing, managing, synchronizing, and terminating sessions between applications. It also provides dialogue control and synchronization during communication.

6. Presentation Layer:

It is responsible for data translation, formatting, compression/decompression, and encryption/decryption. It provides a common format for representing data so that different systems can understand and exchange it.

7. Application Layer:

It is the highest layer of the OSI model. It provides network services directly to user applications such as web browsers, e-mail clients, and file-transfer applications. Examples of Application Layer protocols include HTTP, HTTPS, FTP, SMTP, POP3, and DNS.

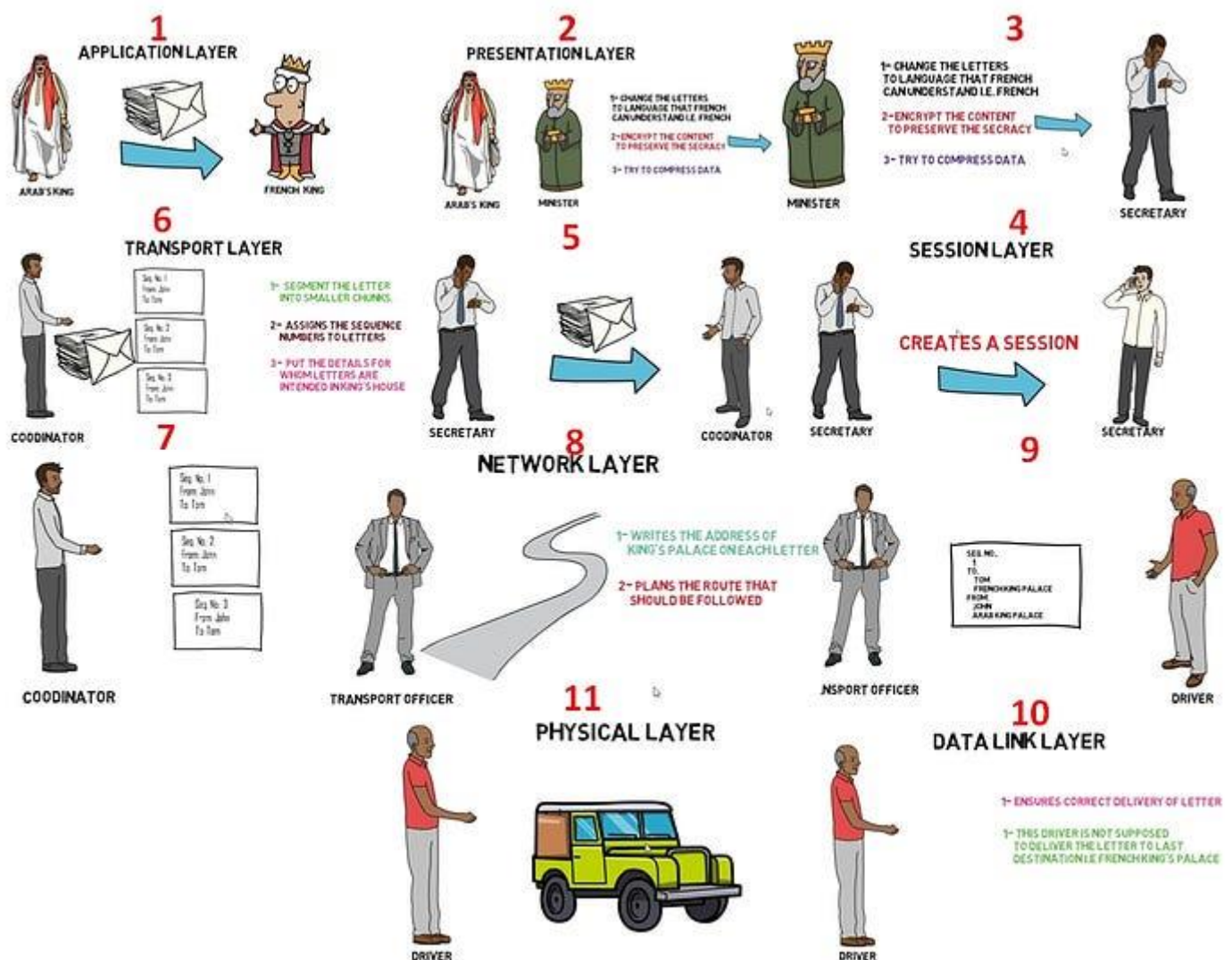


OSI Model Layers	Protocols
Physical layer	ADSL, ISDN, Bluetooth, Ethernet
Data link layer	ARP, MAC, HDLC, STP
Network layer	IP, ICMP, IGMP, RIP, BGP
Transport layer	TCP, UDP, GRE
Session layer	NFS
Presentation layer	SSL
Application layer	SSH, DNS, DHCP, NTP, HTTP

OSI Model Layer	Devices
Physical layer	Hub, repeater
Data link layer	Switch, bridge, Network Interface Card
Network layer	Router, firewall
Transport layer	Gateway
Session layer	Proxy server
Presentation layer	Encryption device
Application layer	Web server, phones

OSI Model Layer	Network Problems
Physical layer	Damaged cables, loose connectors, interference, signal attenuation
Data link layer	Frame errors, collisions, MAC address conflicts
Network layer	IP address conflicts, routing loops, packet loss
Transport layer	Port conflicts, connection timeouts, segment reordering
Session layer	Session hijacking, session termination
Presentation layer	Data corruption, encryption errors
Application layer	Protocol mismatch, service unavailability

Port Number	Protocols
20, 21	File Transfer Protocol (FTP)
22	Secure Shell (SSH)
23	Telnet Protocol
25	Simple Mail Transfer Protocol (SMTP) For Send
53	Domain Name System (DNS)
67, 68	Dynamic Host Configuration Protocol (DHCP)
80	HyperText Transfer Protocol
110/109	Post Office Protocol (POP3) For Mail Receive
137	NetBIOS Name Service
143	Internet Message Access Protocol (IMAP4)
443	Secure HTTP (HTTPS)



APPLICATION LAYER



PRESENTATION LAYER

PRESENTATION LAYER



EBCDIC	ASCII
Space	@
RSP	A
â	B
ã	C
ä	D
å	E
ä	F
â	G
ç	H
ñ	I
ç	J
.	K
<	L
(	M
+	N
	O

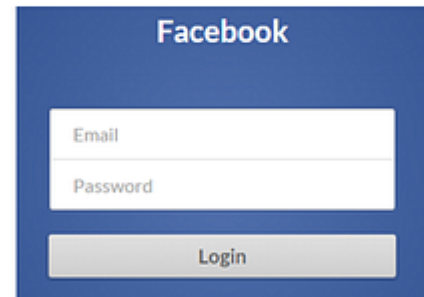
Text to Encrypt

Encrypted Text

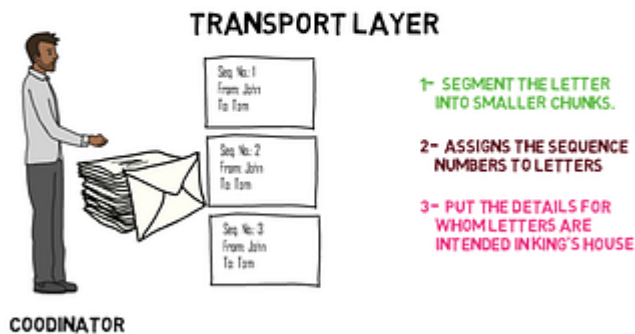
```
2I31bNk8EQPF0N7XYCzdV  
QBUsTem9PdSo5rnfxfuT  
c=
```



SESSION LAYER



TRANSPORT LAYER



Message

Sequence No: 1
Source Port : 80
Destination Port : 80
Add TCP Flags

Segment

Sequence No: 2
Source Port : 80
Destination Port : 80
Add TCP Flags

Segment

Sequence No: 3
Source Port : 80
Destination Port : 80
Add TCP Flags

Segment

1. Breaks message to smaller chunks
2. Assign them sequence number
3. Set the source & destination port Numbers

TCP vs UDP

NETWORK LAYER

NETWORK LAYER



Source IP: 39.51.84.22
Destination IP: 52.69.45.76
Sequence No: 1
Source Port : 80
Destination Port : 80
TCP Flags



Logical addressing: **IP**

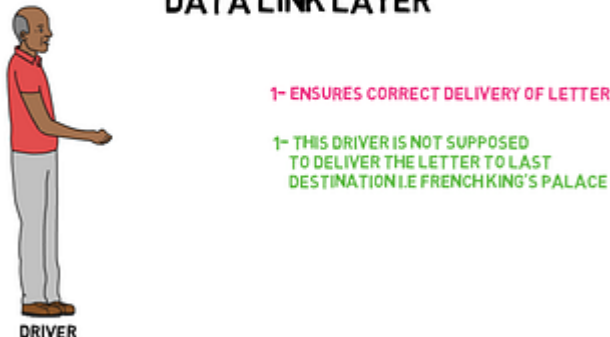
Decides the route to be followed

Routed Protocol: **IP, IPX**

Routing Protocol: **RIP, OSPF**

DATA LINK LAYER

DATA LINK LAYER



Source MAC: 30-65-EC-6F-C4-58
Destination MAC: 00:1B:44:11:3A:B7
Source IP: 39.51.84.22
Destination IP: 52.69.45.76
Sequence No: 1
Source Port : 80
Destination Port : 80
TCP Flags



Error Free delivery

Responsible for next hop delivery

- 1) Logical Link Layer
- 2) Media Access Control

PHYSICAL LAYER



PHYSICAL LAYER



Responsible for 1's
and 0's on wire

Science of voltage,
current and light in
case of fiber



Application Layer

http, ftp, smtp, imap

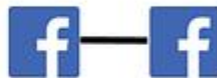
Application level protocols

Presentation Layer



*Format conversion / Encryption /
Compression*

Session Layer



Session Creation

Transport Layer



Application to Application

Network Layer



End to End Delivery

Data Link Layer



Link to Link Delivery

Physical Layer



Actual Hardware

=====HIRA KUMAR=====